



Biology 101: Cellular Respiration and Energy Flow — A Full Reading and Application Session

Biology 101 • Biology

COURSE-OBJ.BIO101.2

Explain how glycolysis, the Krebs cycle, and the electron transport chain each contribute to ATP production, and apply this to real exercise-physiology scenarios. A complete 90-minute session packet with warm-up, guided practice, partner/lab work, independent practice, extension work, assessment, and a full answer key.

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Plan the session

Course and objective

Biology 101 · Biology; COURSE-OBJ.BIO101.2 — Explain the three stages of cellular respiration and how each stage's ATP yield and oxygen requirement connect to a real physiological scenario.

What is included

Cover; instructor guide; key-term and concept preview; original reading or worked examples; warm-up (2 items); guided practice with organizer (4 items); partner/lab discussion (3 items); independent practice (6 items); small-group extension work (4 items); assessment; complete answer key; access and sources page.

Suggested pacing (90-minute session)

Warm-up 8 min · read both passages + vocabulary 15 min · guided stage-by-stage organizer 18 min · partner application discussion 18 min · independent practice 22 min · extension work 5 min · assessment 4 min.

Preparation

Print student pages separately from the answer key, or distribute through a secure LMS. Review the reading/examples and organizer before class. No paid textbook or app is required.

Student directions

Read or work through the material carefully. Support every answer with evidence, work shown, or clear reasoning.



Before you begin

Key terms

ATP — the molecule cells use as usable chemical energy

glycolysis — the oxygen-free first stage, splitting glucose in the cytoplasm

Krebs cycle — the second stage, generating electron carriers in the mitochondria

electron transport chain — the third stage; uses oxygen directly, yields most ATP

anaerobic fermentation — a low-yield backup pathway used without enough oxygen

lactic acid — a byproduct of anaerobic fermentation in muscle cells

Concept focus

Cellular respiration converts glucose into usable energy (ATP) in three connected stages. Glycolysis happens in the cytoplasm, needs no oxygen, and produces a small, quick yield of 2 ATP. The Krebs cycle happens in the mitochondrial matrix, requires oxygen indirectly (it only proceeds if the electron transport chain is running), and produces 2 more ATP plus electron carriers. The electron transport chain, in the inner mitochondrial membrane, uses oxygen directly as the final electron acceptor and produces the large majority of ATP — roughly 32-34 per glucose. When oxygen is scarce, cells fall back on anaerobic fermentation after glycolysis, which is fast but yields far less energy and produces lactic acid as a byproduct.

Model 1 — Comparing the ATP yield of each stage

Glycolysis yields 2 ATP (no oxygen needed); the Krebs cycle yields 2 more ATP (needs oxygen indirectly); the electron transport chain yields roughly 32-34 ATP (needs oxygen directly). The electron transport chain alone accounts for the large majority of total energy from one glucose molecule.

Model 2 — Reading an oxygen-availability scenario

During an all-out 100-meter sprint, oxygen delivery to muscle cells cannot keep pace with demand, so cells rely heavily on glycolysis followed by anaerobic fermentation — fast ATP, but a small yield, with lactic acid building up as a result.

WARM-UP

Warm-up

Answer using what you already know or recall from the previous session.

1. Name the three stages of cellular respiration in order, and state where in the cell each one occurs.

2. Which stage does not require oxygen at all, and approximately how much ATP does it produce?



Reading the Three Stages: A Step-by-Step Overview

Every cell needs a steady supply of ATP, the molecule cells spend as usable energy, and most of that ATP comes from breaking down glucose through three connected stages.

The first stage, glycolysis, takes place in the cytoplasm, outside the mitochondria entirely, and splits one glucose molecule into two molecules of pyruvate. Glycolysis does not require oxygen and produces a net gain of only 2 ATP — small, but fast, and available even when oxygen is limited.

If oxygen is available, the pyruvate from glycolysis moves into the mitochondria for the second stage, the Krebs cycle (also called the citric acid cycle). Here, each pyruvate is broken down further, releasing carbon dioxide and producing a small additional amount of ATP directly (about 2 total), but more importantly, generating electron carriers (NADH and FADH₂) that will fuel the next stage. The Krebs cycle itself does not use oxygen directly, but it grinds to a halt if oxygen is not available downstream, because it depends on the electron transport chain to recycle the electron carriers it needs to keep running.

The third stage, the electron transport chain, takes place across the inner mitochondrial membrane and is where oxygen is used directly, as the final acceptor of the electrons passed along the chain. This stage produces the vast majority of the cell's ATP from one glucose molecule — roughly 32 to 34 ATP, dwarfing the combined total from the first two stages. This is why the presence or absence of oxygen makes such a dramatic difference to how much energy a cell can extract from the same amount of glucose.



Applying the Stages: Sprinting vs. Distance Running

A 100-meter sprinter and a marathon runner both rely on cellular respiration, but their muscles draw on it very differently because of how quickly oxygen can be delivered to working muscle cells. During an all-out sprint, the demand for ATP spikes far faster than the bloodstream can deliver oxygen to muscle tissue, even though the runner is breathing hard. With oxygen effectively unavailable at the rate needed, muscle cells lean almost entirely on glycolysis, and because the pyruvate produced cannot be fully processed through the Krebs cycle and electron transport chain fast enough, cells shift to anaerobic fermentation: pyruvate is converted into lactic acid, allowing glycolysis to keep running quickly, but at a steep cost in total ATP yield per glucose molecule.

A marathon runner's muscles, by contrast, operate at a pace where the cardiovascular system can keep delivering oxygen throughout the race. This allows working muscle cells to run the full three-stage process — glycolysis, the Krebs cycle, and the electron transport chain — continuously, extracting the much larger ATP yield available when oxygen is present. This is a major reason marathon training emphasizes cardiovascular conditioning: the goal is to improve how efficiently oxygen reaches muscle cells, keeping them running the high-yield aerobic pathway for as long as possible rather than being forced into the low-yield anaerobic pathway.

The lactic acid buildup from a sprinter's anaerobic fermentation is also why sprinters feel a burning sensation in their muscles that marathon runners, pacing themselves aerobically, generally avoid until very late in a race, if at all.



Partner / lab discussion

Discuss each item with a partner or lab group before writing a joint response.

1. With a partner, explain why a marathon runner's training emphasizes cardiovascular conditioning, connecting this directly to ATP yield.

2. With a partner, discuss: why do sprinters typically feel a muscle "burn" that marathon runners mostly avoid, according to the passage?

3. Discuss: if a runner's cardiovascular system improved significantly through training, what specific change would you expect in their muscle cells' ATP source during a sustained run, and why?



Independent practice

Work independently. Show your work or cite evidence in every answer.

1. Calculate the approximate total ATP yield from all three stages combined, showing your addition.

2. Explain why the electron transport chain "dwarfs" the ATP contribution of the first two stages combined, using the numbers from the lesson.

3. A student claims "glycolysis is useless because it makes so little ATP." Use evidence from the lesson to argue against this claim.



Independent practice

Work independently. Show your work or cite evidence in every answer.

4. Predict what would happen to a sprinter's ATP production if they tried to run a marathon pace using only the anaerobic pathway the entire time, and explain why this would be unsustainable.

5. Explain the relationship between the Krebs cycle and the electron transport chain using the phrase "electron carriers" in your answer.

6. Design a short original scenario (not sprinting or marathon running) where a cell or organism would rely mainly on anaerobic fermentation, and justify your choice.



Small-group extension work

Use for review or additional application practice.

1. Write a two-to-three sentence explanation of why "more oxygen available" generally means "more ATP produced," using specific stage names.

2. Compare the two passages: what specific piece of evidence from "Applying the Stages" directly illustrates the general fact stated in "Reading the Three Stages" that oxygen absence causes "such a dramatic difference" in ATP yield?



Small-group extension work

Use for learners ready for additional challenge or synthesis.

1. Write an original two-to-three sentence summary explaining cellular respiration to someone who has never taken a biology course, without using the words "mitochondria" or "ATP."

2. Explain why holding your breath briefly during a heavy weightlifting rep does not immediately stop your muscle cells from producing ATP.



Assessment

Complete independently to show what you learned today.

1. Self-assessment: which of the three stages do you feel most confident explaining, and which do you want more practice with?



Assessment · self-assessment

Rate today's work:

4 — I can teach this to someone else.

3 — I can do this independently.

2 — I can do this with help.

1 — I need to revisit this before the next session.



Answer key

1. Glycolysis (cytoplasm), the Krebs cycle (mitochondrial matrix), and the electron transport chain (inner mitochondrial membrane).
2. Glycolysis does not require oxygen and produces a net gain of about 2 ATP.
3. The Krebs cycle itself does not consume oxygen, but it depends on the electron transport chain to recycle its electron carriers (NADH/FADH₂); if oxygen is unavailable for the electron transport chain, the Krebs cycle stops running because it has nowhere to send those carriers.
4. The electron transport chain, yielding roughly 32-34 ATP.
5. The demand for ATP during an all-out sprint spikes faster than the bloodstream can deliver oxygen to the muscle cells, so there is not enough oxygen available at the rate needed to keep the electron transport chain (and the Krebs cycle feeding it) running at that pace.

Accept equivalent, well-supported alternative answers throughout.



Answer key

6. Pyruvate is converted into lactic acid; this allows glycolysis to keep running quickly by recycling the molecules glycolysis needs, even without oxygen, though at a much lower total ATP yield.

7. Better cardiovascular conditioning improves how efficiently oxygen reaches muscle cells, which keeps cells running the high-yield aerobic pathway (all three stages) rather than being forced into the low-yield anaerobic pathway, maximizing ATP produced per glucose molecule.

8. The burning sensation is associated with lactic acid buildup from anaerobic fermentation, which sprinters rely on heavily due to oxygen delivery limits during an all-out effort; marathon runners, pacing aerobically, generally avoid this buildup until very late in a race, if at all.

9. You would expect muscle cells to rely more on the full three-stage aerobic pathway (and less on anaerobic fermentation) for longer, since improved oxygen delivery lets cells keep running the higher-yield electron transport chain instead of switching to the lower-yield anaerobic pathway.

10. 2 (glycolysis) + 2 (Krebs cycle) + roughly 32-34 (electron transport chain) = approximately 36-38 ATP total per glucose molecule.

Accept equivalent, well-supported alternative answers throughout.



Answer key

11. Glycolysis and the Krebs cycle together produce only about 4 ATP directly, while the electron transport chain alone produces roughly 32-34 ATP — more than eight times as much as the first two stages combined.
12. Glycolysis is not useless: it is the only stage that runs without oxygen, making it essential during high-intensity effort (like sprinting) when oxygen cannot be delivered fast enough, and it also supplies the pyruvate that feeds the higher-yield stages when oxygen is available.
13. ATP yield per glucose molecule would stay very low throughout the race, requiring the body to burn through glucose stores far faster for the same energy output, and lactic acid would build up continuously rather than being cleared — this combination would be physically unsustainable over a marathon's distance.
14. The Krebs cycle produces electron carriers (NADH and FADH₂) that carry high-energy electrons to the electron transport chain; the electron transport chain uses those electrons, along with oxygen as the final acceptor, to produce the majority of the cell's ATP.
15. Answers vary; must describe a low-oxygen scenario (e.g., yeast fermenting in a sealed container, muscle cells during a brief maximal lift) and justify it using the oxygen-limitation logic from the lesson.

Accept equivalent, well-supported alternative answers throughout.



Answer key

16. Answers vary; must connect oxygen availability to the electron transport chain running (and therefore the Krebs cycle continuing to feed it), explaining that this combination produces far more ATP than glycolysis and fermentation alone.

17. The sprinter's reliance on glycolysis and anaerobic fermentation (low yield) versus the marathon runner's ability to run the full three-stage aerobic pathway (high yield) directly illustrates the general point that oxygen availability dramatically changes total ATP output.

18. Answers vary; must accurately convey the general concept (cells break down food/sugar in stages to make usable energy, and having enough oxygen lets them make much more of that energy) using accessible language.

19. Muscle cells can still run glycolysis and anaerobic fermentation without a fresh oxygen supply during a brief effort, so ATP production continues at a reduced (anaerobic) rate even while breath is held, though this pathway cannot sustain the same output for long.

20. Answers vary; must name a specific stage (glycolysis, Krebs cycle, or electron transport chain) for each part.

Accept equivalent, well-supported alternative answers throughout.



Access, sources, and license

Extension

Ask for a second method, a counterexample, or a short student-led extension of the concept, with a brief explanation of the choice.

Accommodated seating/504

Follow the individual accommodation plan: extended time, breaks, enlarged print, reduced copying, or an alternate response method as documented.

Additional support

Pre-teach key vocabulary or notation before class. Model the first guided item aloud. Accept oral or dictated responses when appropriate.

Multilingual learners

Preview key terms with examples. Allow brief first-language discussion before independent writing. Provide sentence or solution starters.

Course-level alignment · accessed September 18, 2026

COURSE-OBJ.BIO101.2 — Explain the three stages of cellular respiration and how each stage's ATP yield and oxygen requirement connect to a real physiological scenario. The objective is original wording describing a focused part of the course-level goal.

Course-level learning objective written for this packet; no accreditation, transfer credit, or universal course-equivalency is claimed.

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